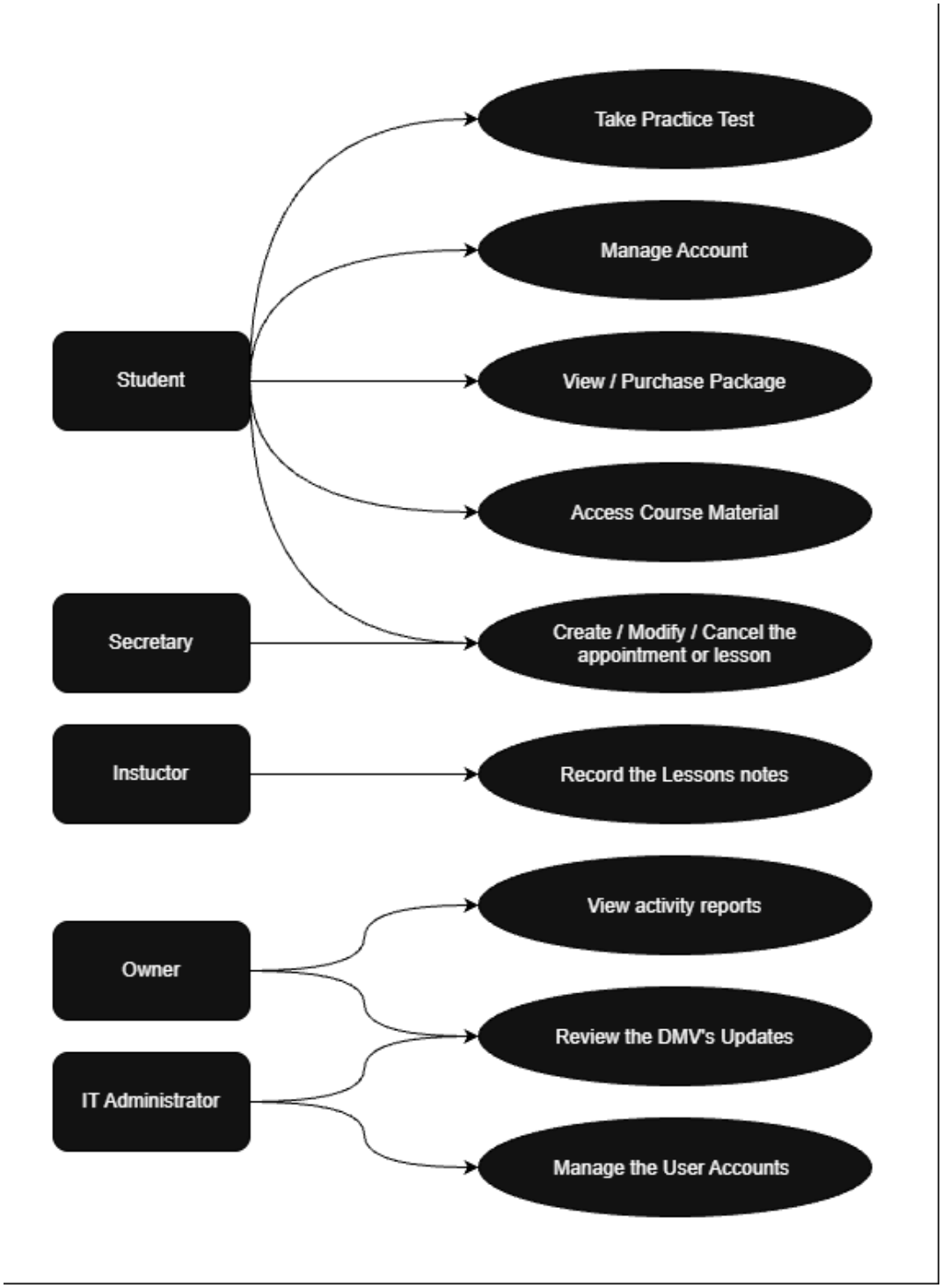
Southern New Hampshire University

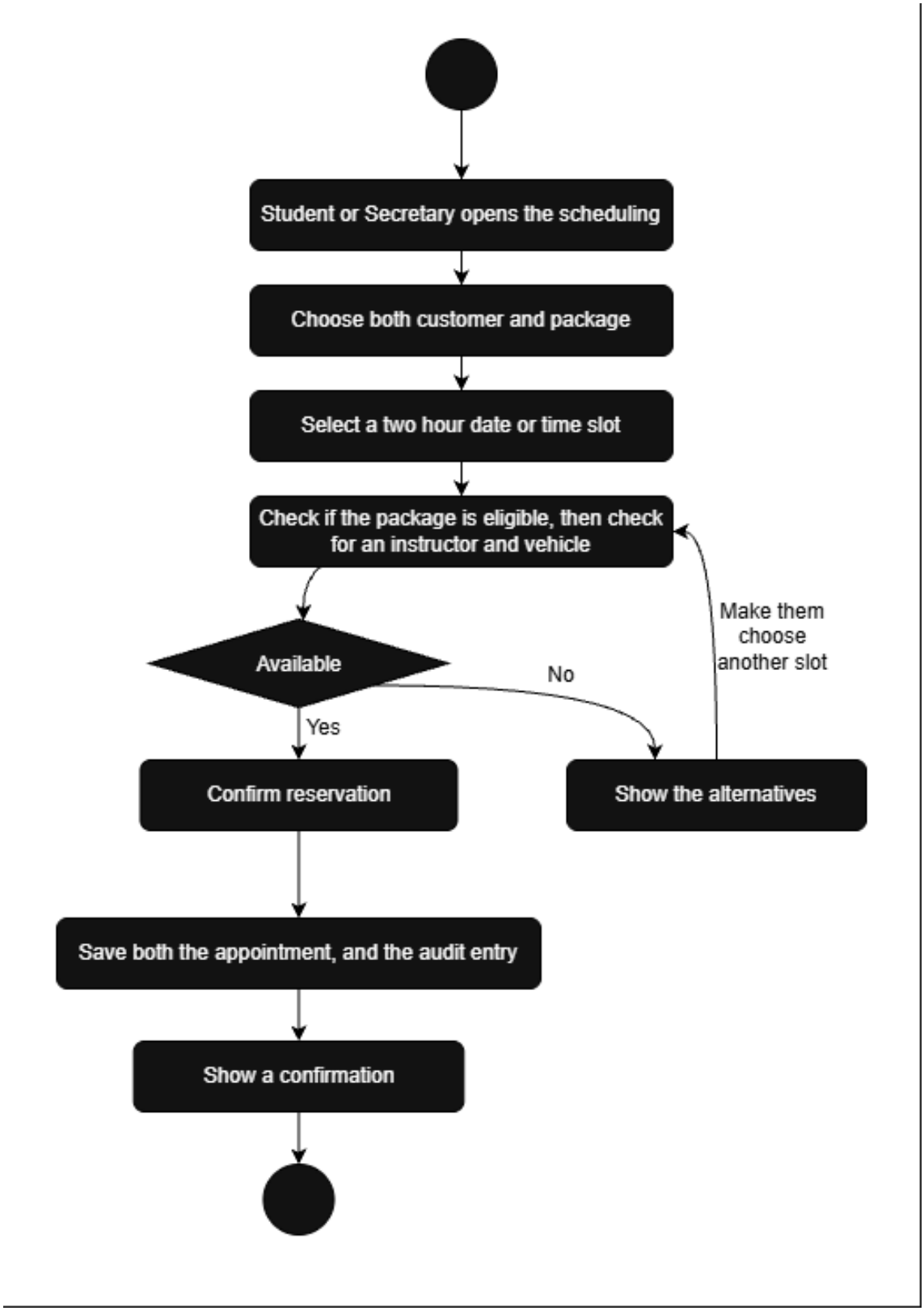
Derick Brown - Project Two - UML Diagrams - CS 255 System Design Document

UML Use Case Diagram



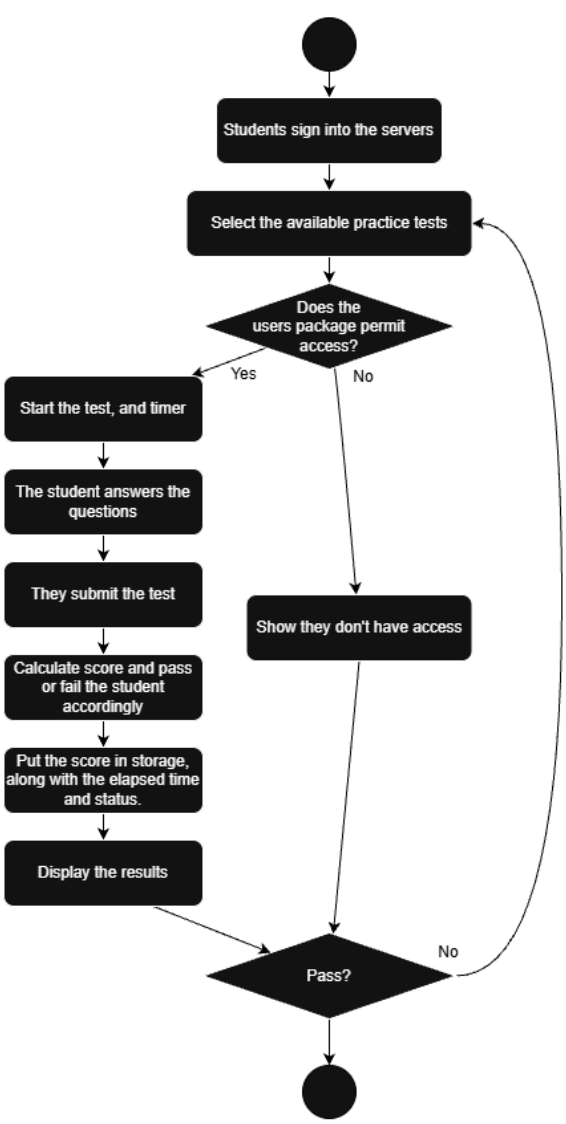
Southern New Hampshire University

UML Activity Diagram

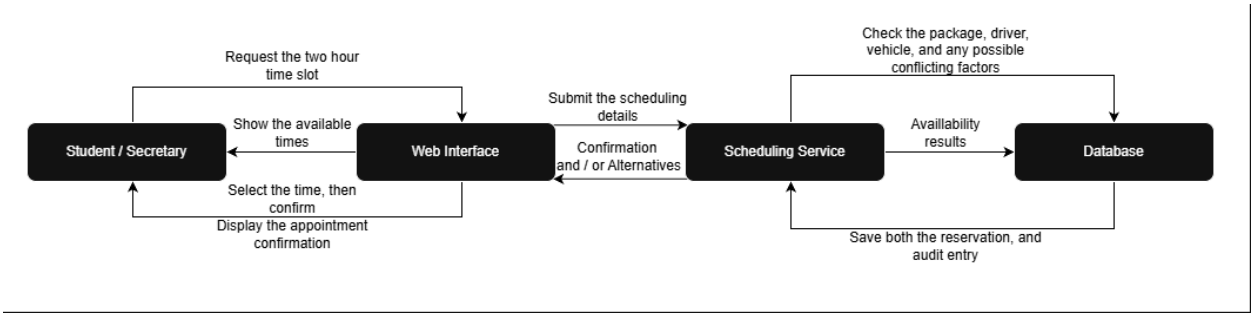


Southern New Hampshire University

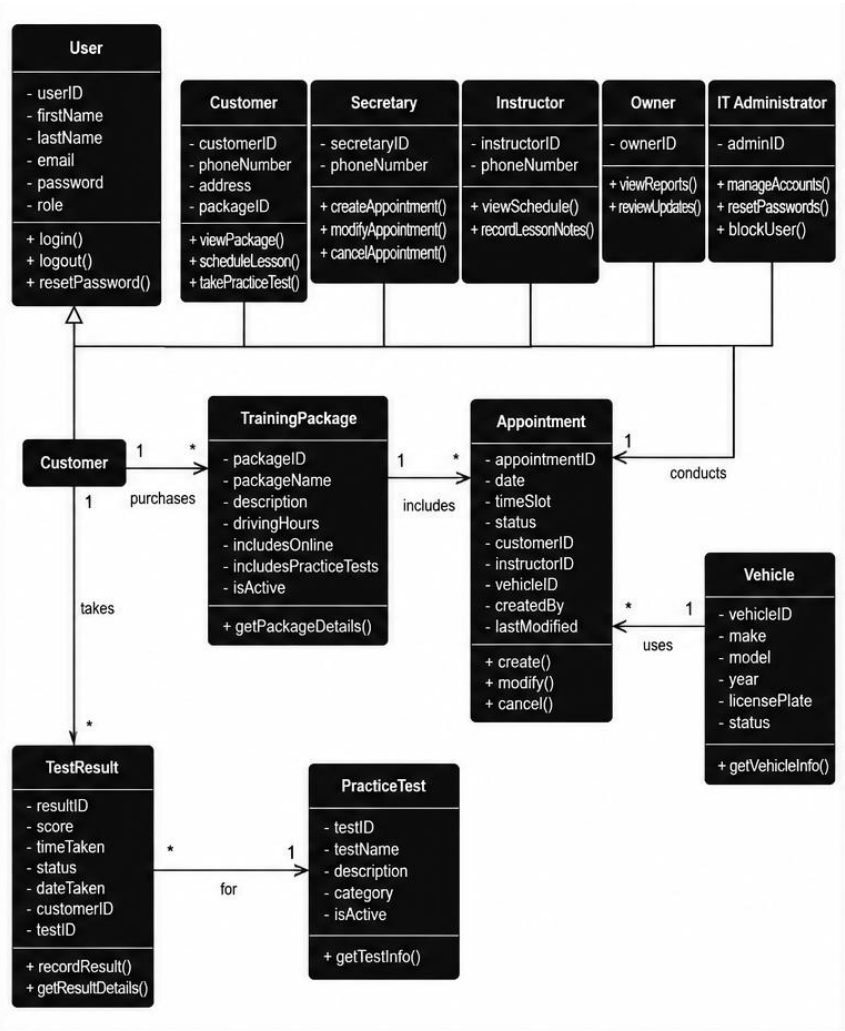
UML Second Activity Diagram



UML Sequence Diagram



UML Class Diagram



Technical Requirements

The technical aspect is broken down into seven tiers of basic requirements, that assume the user meets the requirements in order to properly meet make the most of this system. It's made up in tiers of Hardware and User Devices, Software and Application Architecture, Hosting and Infrastructure, Security and Data Protection, Scheduling and Data Integrity, Learning and Administration, and lastly- Testing, Maintenance, and Unresolved Constraints.

On a hardware level

- **Both students and staff need some kind of internet access on a device capable of reaching it. Driverpass staff themselves needs a reliable and consistent officespace or workstations if they are working remotely. Instructors need an interface that is compatible with mobile to be able to record their lessons, and lesson comments.**

On a Software level

- **Everything in the interfaces, background processes, and generalware of the product needs to be responsive. It should be one of the most focused features, and should synergize and be relatively the same experience throughout whichever device they choose to work on. All of the services should manage everything the service offers. From legitimate eligibility of service, all of the eligibility of the online tests, packages, reservations, and so on.**

On a Hosting level

- **All of the database hosting should be cloud based for immediate ease of access, encrypt all connections properly through HTTPS/TLS, provide backups regularly that also have a process for data recovery. The budget listed for the process should be used properly to account for all auxiliary features.**

On a Security level

- **To maintain security, we should ensure that all users have unique identifiers, password hashing or hiding, role based control, session expiration after a certain duration, and rate limiting for login and entry points. Ability to record appointment scheduling, modification, and cancellations with user IDs and timestamps. We should avoid storing payment card security codes and ensuring that payment processing requires confirmation.**

On a Scheduling / Data level

- **All reservations should record things like the student, the assigned instructor, vehicle, time the test starts, the two hour duration, status, and the creator. We need to check both driver and vehicle availability within the database. Support student self-service and secretary-assisted reservations; preserve history of changed or canceled appointments.**

On a Administration level

- **Store training packages should be configurable records with an active or inactive status and driving-hour privileges. It should track test title, all student attempts, scores, elapsed time, and statuses. Instructors should be able to add lesson comments aswell. The owners export reports, then IT can reset, disable, or restrict accounts. DMV content updates should use an authorized staff review process, automated integration depends on a confirmed DMV source**

On a Maintenance level

- **Test login permissions, scheduling conflicts, cancellation audits, package access, scoring, mobile usability, and backup restoration.**